Fakulteit Natuur- & Landbouwetenskappe

Faculty of Natural & Agricultural Sciences

FSK 176

Fisika vir Ingenieurstudente
Physics for Engineering students

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Denkleiers • Leading Minds • Dikgopolo tša Dihlalefi

FSK 176
Physics for Engineering Students

Department of Physics, University of Pretoria

Version 2015 Revision 1.0

KEEP THIS DOCUMENT HANDY - IT CONTAINS IMPORTANT INFORMATION WHICH YOU WILL NEED TO REFER TO FROM TIME TO TIME

Organisational Component

1 Contact Information

Dr J M Nel
Office 5-11, Natural Science 1 Building (NS1)
Email: fsk176@up.ac.za
Tel: 012 420 3580
Consulting Hours: Check ClickUP or email for an appointment.

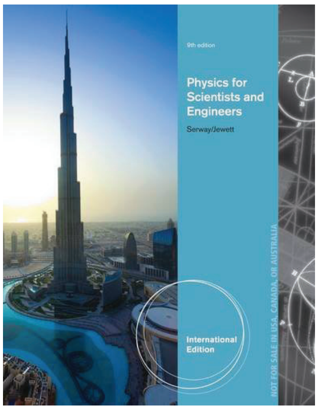
Mrs L Cilliers (Course Administration)
Office 4-16, NS1 Building
Tel: 012 420 4268
Email: fsk176.admin@up.ac.za

Mrs J McDougall (Practical Co-ordinator)
Office 4-17, NS1 Building
Tel: 012 420 4964
Email: fsk176.prac@up.ac.za

2 Prescribed Textbook

It is essential that you have access to a copy of the prescribed textbook as it will be referred to during the lectures as well as all tutorial problems will be assigned from it. The prescribed textbook is:

R.A. Serway and J.W. Jewett, *Physics for Scientists and Engineers with Modern Physics*, Published by Brooks/Cole (Cengage), 2014 9th International Edition (ISBN-13: 978-1-133-95399-9).



The textbook also contains a number of *Examples*, some of which will be discussed in the lectures. Carefully work through the *Examples* not covered during the lectures.

3 Lecture Times

Theory: 4 Lectures per week, presented in English or Afrikaans. Refer to the University of Pretoria 2015 Timetable book for times and venues.

Practicals and Tutorials One 3 hour session is scheduled each week. This session is split between practical session (90 minutes) and tutorial session (90 minutes). See the University of Pretoria 2015 Timetable book for times allocated to your course. Venues and practical groups will be posted on ClickUP.

4 ClickUP

ClickUP is our primary method of communicating with you. Please make sure that you login to ClickUP several times a week. All important information, such as occassional extra class notes, tutorial and homework problems, marks, practical schedules, practical group allocations, test dates and venues etc. will be posted on ClickUP.

5 Final Marks

Final marks will be published by the faculty administration and will be available on your UP Portal page. Do not use "Print/View Academic record" to obtain your results. The Physics Department may not release any final marks to students.

6 Important Dates

Lectures Start:	20 July 2015
Tutorials Start:	Week of 27 July 2015
Practicals Start:	Week of 27 July 2015
Online Tests:	Dates will be announced on ClickUP and in class
Semester Tests:	22 - 29 Aug 2015, and 12 - 17 Oct and 24 Oct 2015
Practical Test:	2 November 2015 (provisional)
Aegrotat (Sick) Test:	29 October 2015 (provisional)
Lectures end:	5 November 2015
Examination:	13 November 2015 (provisional)

7 Assessment Policy

7.1 General

Physics cannot be mastered in a few days of cramming. In order to encourage students to study throughout the semester, a strong emphasis is placed on continuous assessment. For this reason the Semester mark (made up of tests, tutorials and practicals during the semester) makes up half of the final mark.

7.2 Tests and Examinations

NO PROGRAMMABLE CALCULATORS MAY BE USED DURING TESTS OR EXAMS.

- **Online Tests:** Online tests will be available on ClickUP during the semester. An announcement will be placed informing you of the period during which it is available. It will form part of your tutorial mark which will be used when calculating your Semester Mark. These tests will be available after a significant section of work is finsihed.

- **Tutorial Tests:** Tutorial tests will be written at the END of each tutorial session. These tests will be used in the calculation of your Semester mark.

- **Class Tests:** There will not be class tests, since these will be replaced by online tests.

- **Semester Tests:** These tests will cover a larger amount of work than the online tests and will be held during the allocated Test Weeks which are scheduled in the timetable. Venues and times will be announced on ClickUP.

- **Practical Test:** A practical tests will be written at the end of the semester and will cover the work done during the practical sessions. You will be given the data, and will then have to draw a graph from which you will have to calculate certain values. You will not have to do the actual measurements. Venues and times will be announced on ClickUP.

- **Aegrotat (Sick) Tests:** Should you have been ill for any of the tests listed above, you will need to print out and fill in the form "Absent from Test / Afwesig van Toets" (on ClickUP under "Forms/Vorms") and hand it in to Mrs Cilliers on your first day back along with a medical certificate. **Only medical certificates that comply with the university regulations will be accepted. (Doctors need to have been consulted on the day of the test or a day or two before NOT THE DAY AFTER THE TEST!!)**

  The following guidelines need to be followed:

  Tutorial Tests: The average of the remaining tutorial tests will be taken. You are however, encouraged to make an arrangement to reschedule the test with your tutor if possible.

  Weekly Practicals: If you have missed a practical, you **MUST** register for (See "Absent from Practical / Afwesig van Praktikum" under "Forms/Vorms" on ClickUP) and attend one of the catch-up labs. **You have to to register for these sessions.** Report to Mrs McDougall within 7 days in order to register. **Failure to register will result in you receiving 0 (Zero) for the specific practical.**

  Online Tests: There is no sick test for the online tests. A valid doctor's certificate will ensure that the specific test will not be taken into account for the semester mark. You will still need to print out and fill in the form "Absent from Test / Afwesig van Toets" (on ClickUP under "Forms/Vorms") and hand it in to Mrs Cilliers on your first day back. During the semester you will have to complete at least one Online Test, or else receive zero for this component of your semester mark.

  Semester Tests: You will need to fill in the form mentioned above to register for the sick test. One aegrotat test will be taken during the last week of the semester and this will cover ALL the work covered during the semester. This will be the test for all students who have missed a semester test during the semester. Please register your absence from the semester test as soon as you return to class by handing in the form mentioned above. The date and venue will be confirmed in class and on the ClickUP calendar.

  Examination: Should you be ill for the examination, you have to apply for an aegrotat examination via the **faculty administration not the Physics Department**.

- **Examinations:** NB: In order to write the examination, the following requirements need to be met:

  1. Semester mark of minimum 40%
  2. **Final practical mark of a minimum of 50%**
  3. Attendance of 90% of the practicals and tutorials

  Please see the regulation Eng. 3 (b) in the Yearbook of the Engineering Faculty.

- **Aegrotat (Sick) and Supplementary Examinations:** Should you have been ill for the examination, you have to apply for an aegrotat examination via the faculty administration.

  Supplementary examinations are granted when:

  1. the final mark is between 40% and 49%,
  2. OR a final mark of $\geq$ 50% is achieved BUT the required subminimum of 40% has not been achieved in the examination.

  The supplementary exam will be written on the date published by the university in the supplementary exam timetable.

  In the case of a supplementary examination, the final mark is calculated taking the semester mark into account for this second semester course. **NB:** The maximum final mark which can be awarded after a supplementary examination is 50%.

  Please see the regulation Eng. 3 (e) in the Yearbook of the Engineering Faculty.

7.3 Pass Mark

A FINAL average of 50% AND a minimum of 40% for the examination must be obtained in order to pass this module. **NB:** Even if a student obtains 50% or more for the final mark, BUT obtains less than 40% for the examination, that student will not pass the module. **NB:** A student will pass the module with distinction if the final mark is $\geq$ 75%.

Please see the regulation Eng. 3 (c) in the Yearbook of the Engineering Faculty.

7.4 Grading and administrative errors

Please ensure that the marks allocated on your test are correct and that the **correct marks** appear on ClickUP. Should there be any discrepancies, please complete the form "Semester test remark / Vorm: Hermerk van 'n semestertoets" (on ClickUP under "Forms/Vorms") and hand it in along with your test paper in Room 4-17, NS1 within 7 days. **NO CHANGES WILL BE MADE ONE WEEK AFTER THE GRADED PAPERS HAVE BEEN HANDED OUT.**

7.5 Cheat Sheet

During semester tests and exams, NO FORMULA sheet will be handed out with the question paper. It will be your responsibility to draw up your own formula sheet — your Cheat Sheet.
Rules for the Cheat Sheet:

- It must be on the official Cheat Sheet for the specific test/exam
- It must be hand written.
- It must be written in colour any colour but NOT BLACK.
- It may NOT have derivations or worked out solutions to specific problems.
- It must be handed in with your test. Should you want it back afterwards, please staple it to your paper (with your own staple and stapler).
- The page may not be folded.
- You will receive an additional 2% for handing in your cheat sheet. If you do not, however, hand in your cheat sheet, 10% will be deducted from your test mark.
- You may only have ONE cheat sheet in the exam/test venue.

7.6 Barcodes

You will receive a page of sticky label barcodes which you must stick on all your tests, practical book, forms etc. If you misplace your sheet, you will have to request a replacement sheet at a cost of R5. Keep your sheet in a safe place where it will not get worn/torn. If you do not stick your barcode onto your test, you will be penalised.

7.7 Practical Course

ATTENDANCE OF PRACTICAL SESSIONS IS COMPULSORY

The allocated groups will be announced on ClickUP. Please consult Mrs Cilliers with regards to errors in practical time allocations.

Should you not be on the practical list please inform Mrs Cilliers immediately! EVERYONE has to attend the practical sessions.

You have to attend all practical sessions. If you have missed a session with a valid excuse (there are not many of these), you need to register for a catch-up lab with the practical co-ordinators (Mrs McDougall/Mrs Cilliers). A maximum of two make-up sessions per semester will be held.

Students repeating the FSK176 or FSK116 course, **may apply for exemption** from the practical course.

Successful applicants will be notified, but please note that **successful applicants still have to attend the compulsory tutorial sessions.** Do not assume that your application for exemption was successful – please check the list.

Please note: A practical mark of at least 50% and attendance of at least 90% of practical sessions are required for examination entrance.

- **Practical Marks:** After completion of the experiment, the assistant will mark the work in your practical book each week.
- **Practical Test:** Towards the end of the semester, you will be given a practical test on the laboratory work.

7.8 Homework and Tutorial Sessions

Tutorial sessions are compulsory for **ALL** students. **No exemption will be given for the tutorials.** The marks received for the tutorial tests all contribute to your final mark. Should problems occur with regards to attending tutor sessions, please speak to your tutor to reschedule.

During the week, you will receive homework problems which will be similar to the problems which must be solved during the weekly tutorial test. Tutorial tests will be written at the end of each tutorial session. During the session, the tutor will be available to guide and assist students with the homework problems which presented problems. This does not mean that the tutor will be providing all the solutions to the problems; the problems are there for students to solve, and the tutor is merely a guide and hone your problem solving techniques. **DO NOT CONSULT SOLUTIONS BEFORE YOU HAVE ATTEMPTED THE PROBLEM. THIS WILL NOT IMPROVE YOUR PROBLEM SOLVING SKILLS.**

8 Calculation of Marks

Semester Test 1	25%
Semester Test 2	30%
Online Test Average	10%
Weekly Tutorials	15%
Weekly Practicals	12%
Practical Test	8%
Semester Mark	**100%**

Final Mark $\equiv$ (SEM MARK + EXAM MARK)/2

Exemption from practicals: Should you have practical exemption, the calculation of your semester mark is the same as above, with the exception that the total mark (now out of 80) will now be converted to a percentage.

Study Component

9 Main Outcomes

At the end of this module you should be able to apply the principles learnt in the course to analyze and solve relevant problems in: Kinematics, Mechanics, Oscillations, Waves and Thermodynamics.

10 Assessment Criteria

Assessment will focus mainly on the solving of problems, on a similar level to those given as homework exercises. Marks will be allocated for clear, logical reasoning as well as for appropriate diagrams. Demonstrate clearly that you understand the physics that you use to answer the question. This means that a numerically correct answer will not necessarily be awarded full marks.

Knowledge of the theoretical aspect (e.g. derivations) will also be assessed. Here strong emphasis will be placed on clear, logical reasoning.

It is essential that correct units are given in answers.

11 Summary of Chapters Covered

The following summary is based on the prescribed textbook by Serway and Jerwett. Please see ClickUP for more details on each section.

Topic	Chapter	Main Concepts
Measurement	1	Physical quantities, Models, Dimensional Analysis, Units, Order of magnitude, Significant figures
Motion in 1-dimension	2, 13	Position, Velocity, Acceleration, Kinematic equations under constant acceleration, Motion diagrams
Vectors	3	Coordinate systems, Vector and scalar quantities, Components of a vector, Unit vectors
Motion in 2-dimensions	4	Two-dimensional motion with constant acceleration, Projectile motion, Uniform circular motion, Relative velocity and acceleration
Laws of motion	5	Force, Newton's laws, Inertia, Mass, Gravitational force and weight, Frictional forces
Circular motion and resistive forces	6	Uniform circular motion (extended), Motion with resistive forces
Energy	7, 13	Systems and environments, Work done by constant and variable forces, Kinetic enegy-work theorem, Potential Energy, Conservative forces
Conservation of energy	8	Isolated and non-isolated systems, Power
Linear momentum	9	Linear momentum for isolated and non-isolated systems, Collisions in 1- and 2-dimensions, Center of mass, Systems of particles
Rotation of a rigid object	10	Angular position, velocity, and acceleration, Constant angular acceleration, Torque, Moments of inertia, Rotational kinetic energy
Angular momentum	11	Angular momentum for rotating objects, isolated and non-isolated systems
Oscillatory motion	15	Motion of objects connected to springs, Simple harmonic motion, Energy of SHM, Pendulum
Wave motion	16	Propagation of a wave, Traveling waves, Speed of waves on strings, Reflection and transmission
Superposition and standing waves	18	Interference, Standing waves, Boundary conditions, Air columns, Beats
Tempearture	19	Zeroth law, Temperature scales, Thermometers, Thermal expansion of solids and liquids
First law of thermodynamics	20	Heat, Internal energy, Specific heat, Calorimetry, Latent heat, Work and heat in thermodynamic processes, First law, Applications